

Customer Churn Prediction – Data Preprocessing & Feature Engineering

1. Project Overview

This project focuses on preprocessing customer churn data and preparing it for machine learning.

The main objectives are:

- Clean and validate the customer churn dataset.
- Handle categorical and numerical features.
- Apply different encoding techniques.
- Scale numerical features.
- Detect and handle outliers.
- Create new meaningful features.
- Build a complete preprocessing pipeline.

The final processed dataset can be used for customer churn prediction.

2. Dataset

The project uses the `customer\_churn.csv` dataset.

The dataset contains customer information such as:

- Customer ID
- Tenure
- Monthly Charges
- Total Charges
- Contract
- Payment Method
- Paperless Billing
- Senior Citizen
- Churn

The dataset contains 500 rows and 4 primary customer-data columns as provided for the project.

3. Setup Instructions

Install the required Python libraries using:

```
pip install -r requirements.txt
```

Main libraries used:

- pandas
- numpy
- scikit-learn
- matplotlib
- seaborn

Run the preprocessing script using:

```
python main.py
```

4. Data Exploration

The dataset was loaded using Pandas and checked for:

- Dataset shape
- Column names
- Data types
- Missing values
- Duplicate records
- Numerical and categorical features
- Churn distribution

These checks helped identify the structure and quality of the dataset before preprocessing.

5. Data Cleaning

The following cleaning operations were performed:

Missing Values

Missing values were identified and handled appropriately so that they do not affect the preprocessing pipeline.

Duplicate Records

Duplicate records were checked and removed where necessary.

Data Types

Columns were converted to appropriate numerical or categorical formats before applying transformations.

Data Validation

Required columns and dataset structure were validated before continuing with preprocessing.

6. Categorical Encoding

Categorical variables cannot be directly used by most machine learning algorithms, so encoding techniques were applied.

The project implements multiple encoding approaches:

One-Hot Encoding

One-hot encoding converts categorical values into separate binary columns.

Example:

Contract

Month-to-month

One year

Two year

can be transformed into separate numerical indicator columns.

Label Encoding

Label encoding assigns numerical labels to categorical values where an ordered numerical representation is appropriate.

Additional Categorical Transformation

Categorical features were transformed into machine-learning-compatible numerical representations as part of the preprocessing workflow.

7. Feature Scaling

Numerical features were scaled to make their ranges more suitable for machine learning.

Two scaling techniques were considered/applied:

Standard Scaling

StandardScaler transforms numerical values based on their mean and standard deviation.

Min-Max Scaling

MinMaxScaler transforms values into a fixed range, generally between 0 and 1.

Scaling helps prevent features with larger numerical ranges from dominating the model.

8. Outlier Detection

Outliers were identified using statistical methods.

The Interquartile Range (IQR) method was used:

Q1 = 25th percentile

Q3 = 75th percentile

$IQR = Q3 - Q1$

Values outside the acceptable IQR range were treated as potential outliers.

Outlier handling helps improve the reliability of the processed dataset.

9. Feature Engineering

New features were created from existing customer information to improve the predictive value of the dataset.

Examples include:

- Customer Lifetime Value
- Payment Efficiency
- Tenure-related features
- Monthly charge based derived features
- Other meaningful combinations of existing variables

Feature engineering helps represent customer behaviour in a more useful form for machine learning.

10. Feature Selection

Relevant numerical and categorical features were reviewed to identify the variables most useful for predicting customer churn.

Correlation and feature importance techniques can be used to understand the relationship between features and the churn target.

Irrelevant or redundant information should be excluded from the final model input.

11. Preprocessing Pipeline

A complete preprocessing workflow was created to keep the transformations consistent.

The pipeline includes:

Raw Customer Data



Data Validation



Data Cleaning



Missing Value Handling



Categorical Encoding



Feature Scaling



Outlier Handling



Feature Engineering



Feature Selection



Processed Data

Using a pipeline ensures that the same preprocessing steps can be applied consistently to new customer data.

12. Technical Requirements

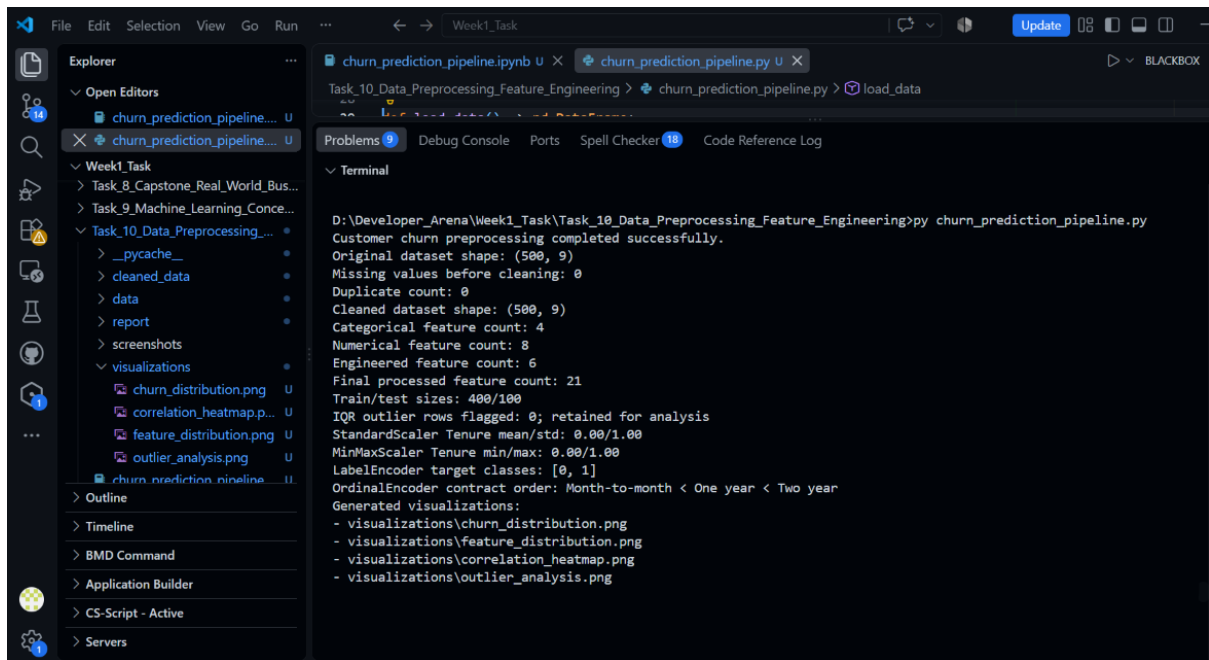
The project satisfies the required technical objectives:

- Implemented multiple categorical encoding techniques.
- Applied numerical feature scaling.
- Performed outlier detection and handling.
- Created more than five engineered features.
- Implemented a complete preprocessing workflow.
- Documented the reasoning behind preprocessing steps.

13. Project Structure

Task_10_Data_Preprocessing_Feature_Engineering/

```
|
|
|— data/
|   └─ customer_churn.csv
|
|
|— report/
|   └─ preprocessing_report.md
|
|
|— screenshots/
|
|
|— visualizations/
|
|
|— churn_prediction_pipeline.ipynb
|— main.py
|— requirements.txt
└─ README.md
```



14. Testing and Validation

The preprocessing workflow was tested to verify that:

- The dataset loads correctly.
- Required columns are available.
- Missing values are handled.
- Categorical features are encoded.
- Numerical features are scaled.
- Engineered features are generated.
- The complete preprocessing pipeline executes successfully.

15. Conclusion

The project demonstrates a complete data preprocessing and feature engineering workflow for customer churn prediction.

The raw customer data was cleaned, transformed, encoded, scaled, and enhanced with new features. The final preprocessing pipeline provides machine-learning-ready data and can be used as the foundation for a customer churn prediction model.